

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

MAY EXAMINATION

UCCC3073 / UCCG3073 DATA SCIENCE

TUESDAY, 27 MAY 2025

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF SCIENCE (HONOURS)
STATISTICAL COMPUTING AND OPERATIONS RESEARCH

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the **FIRST (1)** question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

UCCC3073 / UCCG3073 DATA SCIENCE**SECTION A (Answer ALL questions)**

- Q1. (a) Complete BookList Ltd. is a small business based in Kampar that offers a wide range of books for sale. As the customer base expands, the company needs a database system to efficiently manage its information. Below are the data the company wishes to store in the database.

Books	(bookID, title, ISBN, genre, price, publicationDate, stockQuantity, publisher, pageCount, description, coverImageLink)
Authors	(authorID, authorName, biography, nationality, dateOfBirth, AuthorPhoneNo, authorEmail)
Customers	(customerID, customerName, customerEmail, customerPhoNo, customerAddress, username, passwordHash, registrationDate, loyaltyPoints, preferredLanguage)
Orders	(orderID, customerID, orderDate, orderStatus, shippingAddress, totalPrice, paymentStatus, paymentMethod, shippingMethod, trackingNo)
OrderDetails	(orderDetailID, orderID, bookID, quantity, priceAtOrder, discount, subtotal)
BooksAuthors	(bookID, authorID)

- (i) Plot a conceptual model to illustrate the relationships between entities. (6 marks)
 - (ii) Name the type of database that you will construct for the company. (1 mark)
 - (iii) Name and describe the principle that you will adopt for the database of Q1(a)(ii). (8 marks)
- (b) Consider the following 3 vectors, A, B and C.
- A = (1, 3, 5)
 B = (2, 5, 6)
 C = (3, 5, 6)
- (i) Calculate the cosine similarities between vector C and the other 2 vectors. Round your answer to the nearest 3 decimal points. (9 marks)
 - (ii) Using the answer of Q1(b)(i), determine which vector is most similar to vector C. (1 mark)
- [Total : 25 marks]

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- Q2. A pharmaceutical company asserts that the average recovery time for patients using their new medication for a specific condition differs from the current average of 10 days. To evaluate this claim, a clinical trial was conducted using a random sample of 7 patients, where the average recovery time for those on the new medication was 9.14 days with a standard deviation of 2 days.

Conduct a hypothesis test to determine if there is sufficient evidence to support the company's claim at a 5% significance level.

- (a) State the null and alternative hypotheses. (2 marks)
- (b) Determine the following:
- (i) Type of statistic test to be performed. (1 mark)
 - (ii) Test result. Round your answer to the nearest 3 decimal points. (4 marks)
 - (iii) p-value. Round your answer to the nearest 3 decimal points. (2 marks)
- (c) Using the results of Q2(b), justify whether the null hypothesis of Q2(a) can be rejected. (2 marks)
- (d) Assume that the random sample of Q2 is increased to 300 patients and the population standard deviation is 5 days.
- Determine the following:
- (i) Type of statistic test to be performed. (1 mark)
 - (ii) Test result. Round your answer to the nearest 3 decimal points. (4 marks)
 - (iii) p-value. Round your answer to the nearest 3 decimal points. (2 marks)
 - (iv) Upper- and lower-bounds. Round your answer to the nearest 3 decimal points. (4 marks)
- (e) Using the computed results of Q2(d), justify whether the null hypothesis of Q2(a) can be rejected. (3 marks)
- [Total : 25 marks]

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Q3. (a) Given a transaction containing the following 6 items:

1, 2, 3, 4, 5, 6

Given $k = 4$ in k -itemsets,

- (i) Determine the total number of possible itemsets for this transaction. (2 marks)
- (ii) Construct the enumerate tree to show the itemsets in each level. (10 marks)

(b) Figure Q3.1 shows two groups of data, marked with X and O, respectively.

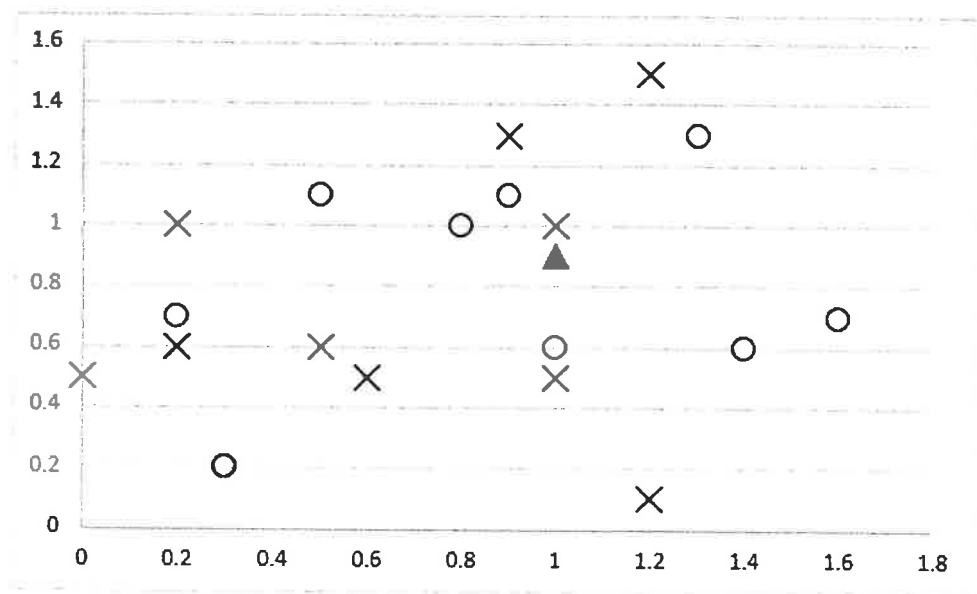


Figure Q3.1

Given a new data point ($\blacktriangle$) in Figure Q3.1,

- (i) Determine the class for the new data point if a 1-NN classification model is used. (2 marks)
 - (ii) Determine the class for the new data point if a 3-NN classification model is used. (2 marks)
 - (iii) Explain how an increase in k would affect the expected loss in a k -NN classification model in terms of Bias and Variance. (9 marks)
- [Total : 25 marks]

UCCC3073 / UCCG3073 DATA SCIENCE**SECTION B (Choose only ONE question)**

Q4. You have developed two different classification models, namely Model A and Model B, for detecting faulty semiconductor chips. Both models were tested using 5000 chip samples from the company. Among these, 500 samples are faulty chips. Model A correctly predicts 450 faulty chips and 3000 working chips. Meanwhile, Model B correctly predicts all the faulty chips and 2900 working chips.

- (a) Describe true positive (TP), true negative (TN), false positive (FP) and false negative (FN) in this context. (4 marks)
- (b) Construct confusion matrix for both Model A and Model B. (4 marks)
- (c) Using the answer of Q4(b), calculate the following evaluation metrics for both models. Identify the superior model for each metric. Round your answer to the nearest 2 decimal points.
 - (i) Classification accuracy rate. (3 mark)
 - (ii) Recall. (3 mark)
 - (iii) Precision. (3 mark)
 - (iv) Specificity. (3 mark)
- (d) Using the computed results of Q4(c), justify which model you would propose to the company. (5 marks)

[Total : 25 marks]

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- Q5. Suppose you have trained two different models, namely Model A and Model B, for diagnosing cancer patients in a private hospital. Model A achieves an accuracy of 90%, while Model B achieves an accuracy of 92%. The performances of these models were evaluated using 2000 patient samples. Among these, 500 samples are cancer samples. Model A correctly predicts 450 cancer samples and 1350 healthy patient samples. Meanwhile, Model B correctly predicts 400 cancer samples and 1450 healthy samples.
- (a) Identify positive and negative classes. (2 marks)
 - (b) Calculate the baseline performance. (2 marks)
 - (c) Discuss how you would compare these models beyond accuracy. In your answer, provide **THREE (3)** other evaluation metrics that you would consider, and explain why might they provide a more complete picture of model performance. (6 marks)
 - (d) Using the answer of Q5(c), calculate the metric values for both models. Identify the superior model for each metric. Round your answer to the nearest 2 decimal points. (9 marks)
 - (e) Using the computed results of Q5(b) and Q5(d), justify which model you would propose to the hospital. (6 marks)
- [Total : 25 marks]
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APPENDIX

<u>Proximity measures</u>	
Euclidean distance	Hamming distance
$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - x)^2 + (y_i - y)^2}$	$d(a, b) = (a \oplus b)$
Manhattan distance	Simple matching coefficient
$d(x, y) = \sum_{i=1}^n x_i - x + y_i - y $	$SMC = \frac{\text{num of matching attriute values}}{\text{num of attributes}}$
Supremum distance	Jaccard coefficient
$d(x, y) = \max(x_i - x , y_i - y)$	$J = \frac{\text{num of matching presences}}{\text{num of attributes not involved in 00 matches}}$
	Cosine similarity
	$\cos(x, y) = \frac{x \cdot y}{ x y }$

<u>Regression</u>		
Straight line	t value	Standard errors of coefficients
$\hat{y} = \beta_0 + \beta_1 x$	$t = \frac{\hat{\beta}}{SE}$	$SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}$
Coefficients	Coefficient of Determination	
$\hat{\beta}_1 = \frac{p \sum xy - \sum x \sum y}{p \sum x^2 - (\sum x)^2}$	$R^2 = \frac{p \sum xy - \sum x \sum y}{\sqrt{(p \sum x^2 - (\sum x)^2) * (p \sum y^2 - (\sum y)^2)}}$	$SE(\hat{\beta}_0) = s \sqrt{\frac{1}{p} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$
$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$	Pearson correlation	
Residuals	$R = \sqrt{R^2}$	
$\hat{\epsilon} = y - \hat{y}$		
$s^2 = \frac{\sum \hat{\epsilon}^2}{p - 2}$		
$s = \sqrt{s^2}$		

<u>Naïve Bayes</u>		
Bayes' rule	Probability density function	Laplace correction
$P(H E) = \frac{P(E H)P(H)}{P(E)}$	$pdf(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\bar{x})^2}{2\sigma^2}}$	$\frac{x_i + 1}{c + \alpha}$

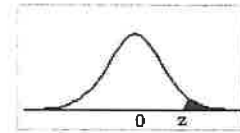
<u>Association Rules</u>		
Association rule	Combination probability	Confidence
$X \rightarrow Y$	${}^nC_r = \frac{n!}{r! (n-r)!}$	$c = \frac{freq(X, Y)}{freq(X)}$
Support		Lift
$s = \frac{freq(X, Y)}{N}$		$l = \frac{Confidence(X, Y)}{Support(X)}$

Evaluation metrics	
Classification accuracy $acc = \frac{TN + TP}{TN + TP + FN + FP}$	Precision / Positive Predictive Value $PPV = \frac{TP}{TP + FP} = 1 - FDR$
Misclassification rate $1 - acc$	F1 score $F_1 = \frac{2TP}{2TP + FP + FN}$
Sensitivity / Recall $recall = \frac{TP}{TP + FN}$	Overall model cost $FP.Cost_{FP} + TP.Cost_{TP}$
Specificity $spec = \frac{TN}{TN + FP}$	Model cost per case $\frac{\text{Overall model cost}}{N}$
False discovery rate $FDR = \frac{FP}{FP + TP}$	Model profit per case $-\text{model cost per case}$

Miscellaneous		
Linear scaling for [a, b] $x' = (b - a) \frac{x - \min(x)}{\max(x) - \min(x)} + a$	Z-test $Z = \frac{\bar{x} - \mu}{SE_{\bar{x}}}$	Population standard deviation $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$
Inverse linear scaling for [a, b] $x = (x' - a) * \frac{\max(x) - \min(x)}{b - a} + \min(x)$	t-test $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$	Sample standard deviation $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$
z-score normalization $x' = \frac{x - \mu}{\sigma}$	Standard error $SE = \frac{\sigma}{\sqrt{n}}$	
Mean, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Confidence intervals $CI = \bar{x} \pm z * SE$	

UCCC3073 / UCCG3073 DATA SCIENCE**Normal Curve Areas (Standard normal probability in right-hand tail)**for negative values of z , areas are found by symmetry confidence level to Z-score

90%	1.645
95%	1.96
98%	2.33
99%	2.575



z	Second decimal place of z									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

UCCC3073 / UCCG3073 DATA SCIENCE**t Distribution: Critical Values of t**

Degree of Freedom	1-tailed α	Significant level						
		0.25	0.2	0.15	0.1	0.05	0.025	0.01
2-tailed α		0.5	0.4	0.3	0.2	0.1	0.05	0.01
1		1.000	1.376	1.963	3.078	6.314	12.706	31.821
2		0.816	1.061	1.386	1.886	2.920	4.303	6.965
3		0.765	0.978	1.250	1.638	2.353	3.182	4.541
4		0.741	0.941	1.190	1.533	2.132	2.776	3.747
5		0.727	0.920	1.156	1.476	2.015	2.571	3.365
6		0.718	0.906	1.134	1.440	1.943	2.447	3.143
7		0.711	0.896	1.119	1.415	1.895	2.365	2.998
8		0.706	0.889	1.108	1.397	1.860	2.306	2.896
9		0.703	0.883	1.100	1.383	1.833	2.262	2.821
10		0.700	0.879	1.093	1.372	1.812	2.228	2.764
11		0.697	0.876	1.088	1.363	1.796	2.201	2.718
12		0.695	0.873	1.083	1.356	1.782	2.179	2.681
13		0.694	0.870	1.079	1.350	1.771	2.160	2.650
14		0.692	0.868	1.076	1.345	1.761	2.145	2.624
15		0.691	0.866	1.074	1.341	1.753	2.131	2.602
16		0.690	0.865	1.071	1.337	1.746	2.120	2.583
17		0.689	0.863	1.069	1.333	1.740	2.110	2.567
18		0.688	0.862	1.067	1.330	1.734	2.101	2.552
19		0.688	0.861	1.066	1.328	1.729	2.093	2.539
20		0.687	0.860	1.064	1.325	1.725	2.086	2.528
21		0.686	0.859	1.063	1.323	1.721	2.080	2.518
22		0.686	0.858	1.061	1.321	1.717	2.074	2.508
23		0.685	0.858	1.060	1.319	1.714	2.069	2.500
24		0.685	0.857	1.059	1.318	1.711	2.064	2.492
25		0.684	0.856	1.058	1.316	1.708	2.060	2.485
26		0.684	0.856	1.058	1.315	1.706	2.056	2.479
27		0.684	0.855	1.057	1.314	1.703	2.052	2.473
28		0.683	0.855	1.056	1.313	1.701	2.048	2.467
29		0.683	0.854	1.055	1.311	1.699	2.045	2.462
30		0.683	0.854	1.055	1.310	1.697	2.042	2.457
31		0.682	0.853	1.054	1.309	1.696	2.040	2.453
32		0.682	0.853	1.054	1.309	1.694	2.037	2.449
33		0.682	0.853	1.053	1.308	1.692	2.035	2.445
34		0.682	0.852	1.052	1.307	1.691	2.032	2.441
35		0.682	0.852	1.052	1.306	1.690	2.030	2.438
36		0.681	0.852	1.052	1.306	1.688	2.028	2.434
37		0.681	0.851	1.051	1.305	1.687	2.026	2.431
38		0.681	0.851	1.051	1.304	1.686	2.024	2.429
39		0.681	0.851	1.050	1.304	1.685	2.023	2.426
40		0.681	0.851	1.050	1.303	1.684	2.021	2.423
41		0.681	0.850	1.050	1.303	1.683	2.020	2.421
42		0.680	0.850	1.049	1.302	1.682	2.018	2.418
43		0.680	0.850	1.049	1.302	1.681	2.017	2.416
44		0.680	0.850	1.049	1.301	1.680	2.015	2.414
45		0.680	0.850	1.049	1.301	1.679	2.014	2.412
46		0.680	0.850	1.048	1.300	1.679	2.013	2.410
47		0.680	0.849	1.048	1.300	1.678	2.012	2.408
48		0.680	0.849	1.048	1.299	1.677	2.011	2.407
49		0.680	0.849	1.048	1.299	1.677	2.010	2.405
50		0.679	0.849	1.047	1.299	1.676	2.009	2.403
60		0.679	0.848	1.045	1.296	1.671	2.000	2.390
70		0.678	0.847	1.044	1.294	1.667	1.994	2.381
80		0.678	0.846	1.043	1.292	1.664	1.990	2.374
90		0.677	0.846	1.042	1.291	1.662	1.987	2.368
100		0.677	0.845	1.042	1.290	1.660	1.984	2.364
150		0.676	0.844	1.040	1.287	1.655	1.976	2.351
200		0.676	0.843	1.039	1.286	1.653	1.972	2.345
250		0.675	0.842	1.036	1.282	1.645	1.960	2.327